$$G = \nabla f(W)$$

$$M = \mu M + G$$

$$\varepsilon = \text{Quantile}(|M|, p)$$
$$O = \text{sign}(M) \cdot \max(|M| - \varepsilon, 0)$$

$$B = M - O$$

$$X_0 = B / (\|B\|_F + \varepsilon)$$

repeat $k = 0 \ldots K-1$:

$$A = X_k X_k^T$$
$$B_k = b \cdot A + c \cdot A^2 \text{ (b, c per shape)}$$
$$X_{k+1} = a \cdot X_k + B_k X_k \text{ (a per shape)}$$

$$W \leftarrow W - \eta X_K$$